

ENGINEERING ECONOMICS

Force Field Analysis

- Force Field Analysis is best carried out in small group of about six to eight people using flip chart paper or overhead transparencies so that everyone can see what is going on.
- The first step is to agree the area of change to be discussed. This might be written as a desired policy goal or objective.
- All the forces in support of the change are then listed in a column to the left (driving the change forward) while all forces working against the change are listed in a column to the right (holding it back).
- The driving and restraining forces should be sorted around common themes and should then be scored according to their 'magnitude', ranging from one (weak) to five (strong).
- The score may well not balance on either side.

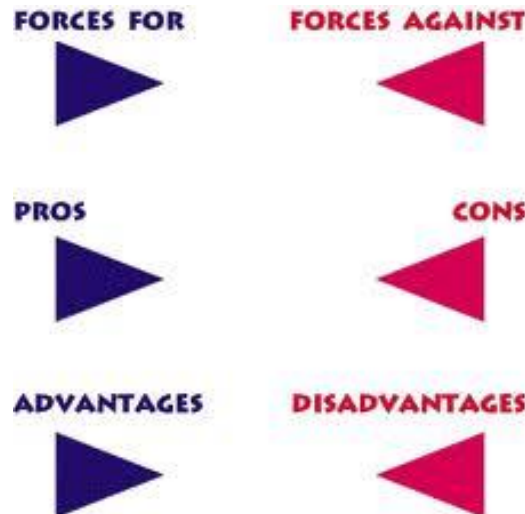
Force Field Analysis

- The basic idea behind the force field analysis is that a certain given situation remains the way it is because of counterbalancing forces, or because of the state of equilibrium between forces that drive change and oppose change.
- In order to promote change, either the driving forces should be strengthened, or the resisting forces should be weakened.

When to use the tool?

The force field analysis is ideally used for diagnosing a problem. You can use it to

1. Analyze the balance of power
2. Identify the key roles involved in decision-making
3. Identify who supports and opposes change within the organization
4. Explore ways to influence those who are against change
5. Decide whether to go ahead with a proposed change or not



FORCE FIELD ANALYSIS

FOR



Sample text

Sample text

Sample text

Sample text

Sample text

AGAINST



Sample text

Sample text

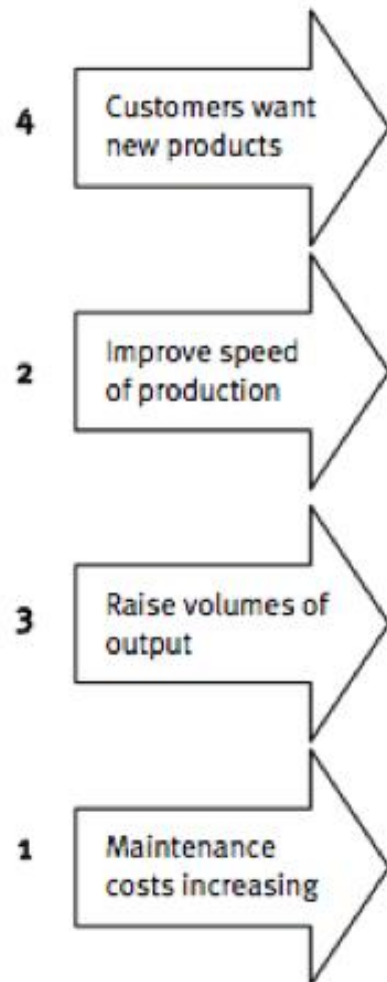
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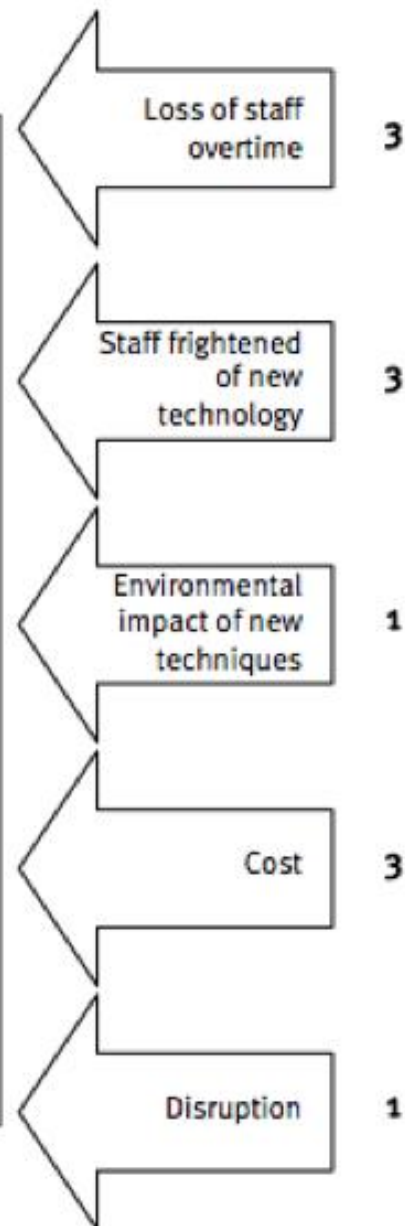
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Click here to add your
text

Forces for change



Total: 10

Forces against change



Total: 11

Plan:

**Upgrade factory
with new
manufacturing
machinery**

Ishikawa Diagram/Fishbone / Cause and Effect Diagram

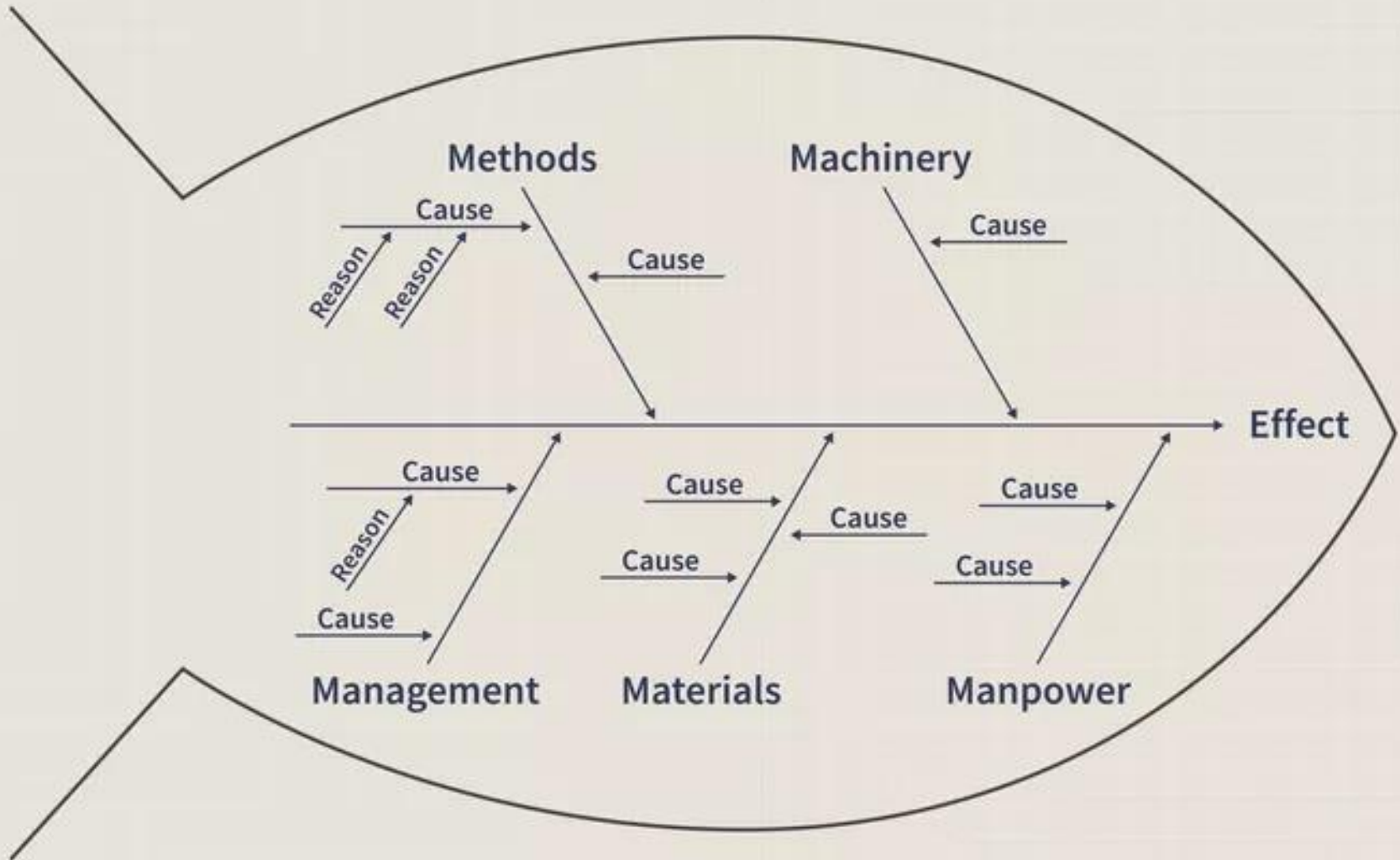
- An Ishikawa diagram is used to show the causal factors that go into some final outcome, often related to a production or design problem.
- They are named after Japanese engineering professor Kaoru Ishikawa in the 1960s, who helped apply them to manufacturing processes.
- Shaped somewhat like a fish, these charts are sometimes called fishbone or 'Fishikawa' diagrams.
- Ishikawa diagrams often follow the "Six M's": manpower, machinery, methods, materials, measurement, and mother nature.

6 Ms

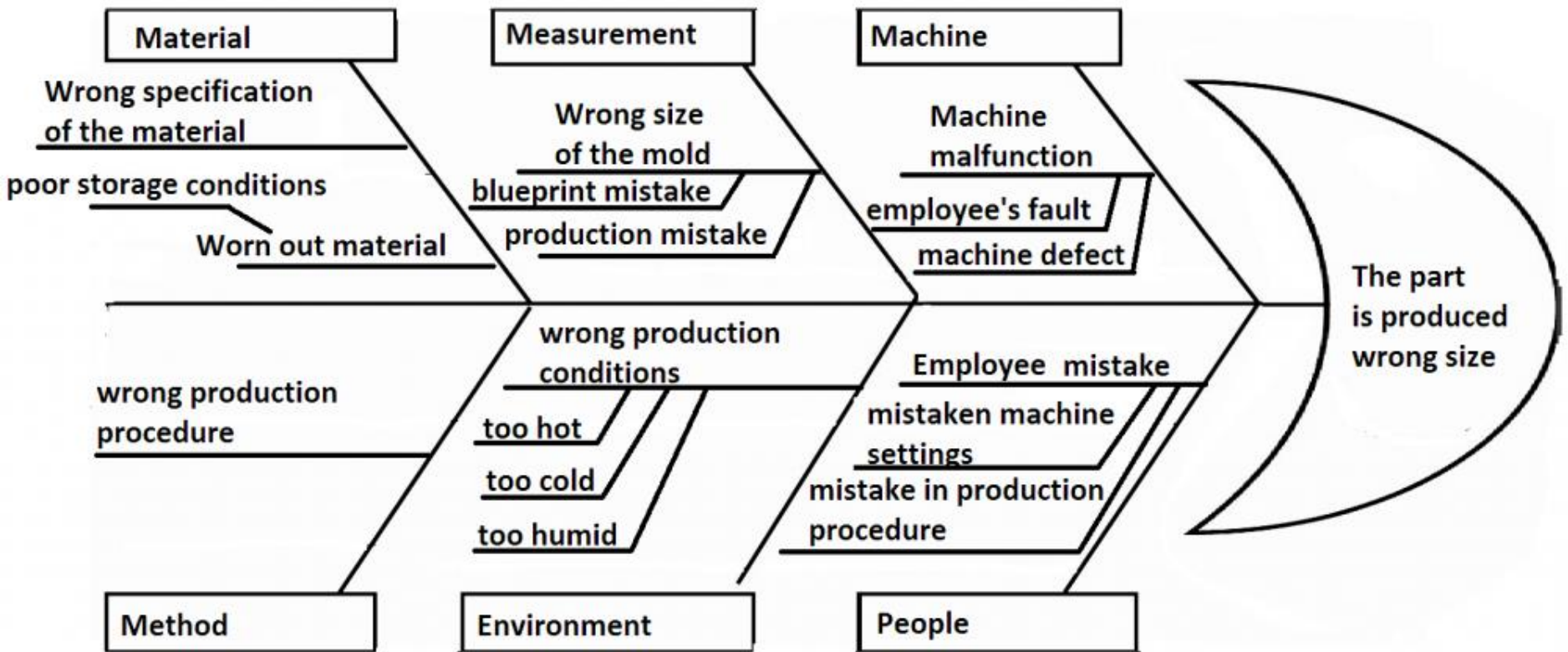
Each of the "bones" or "ribs" in a classic Ishikawa diagram represents an issue relevant to quality control and may be described by the six "M"s, where failures or potential failures can be identified and rectified. These are:

1. **Manpower** - The training, skill, and attitude of the employees or workers
2. **Machines** - Maintenance of machines, whether upgrades to better technology is needed
3. **Materials** - Are raw materials and inputs properly labeled, stored, and of high quality. Have they been ordered in the right size and quantity?
4. **Measurement** - Are methods of measurement and control correct and accurate. Do they need to be adjusted?
5. **Mother Nature** - Often uncontrollable environmental factors like fire or bad weather, but certain safety measures can be undertaken, as well as insurance purchased for damage or disaster
6. **Method** - Does the production process have the most efficient number of steps, are there bottlenecks, is it overly complex and error-prone?

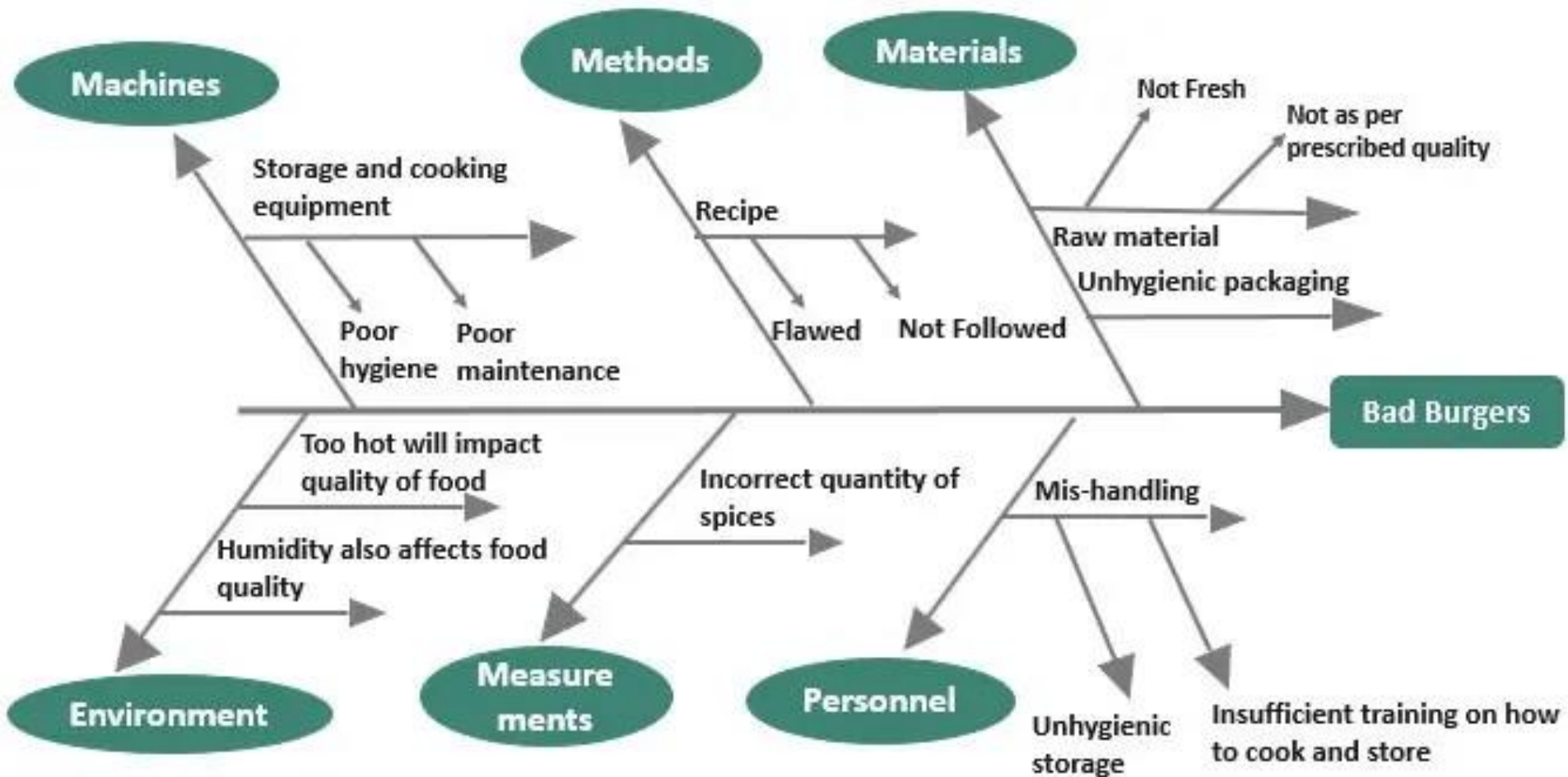
Ishikawa Diagram



Example - Engineering



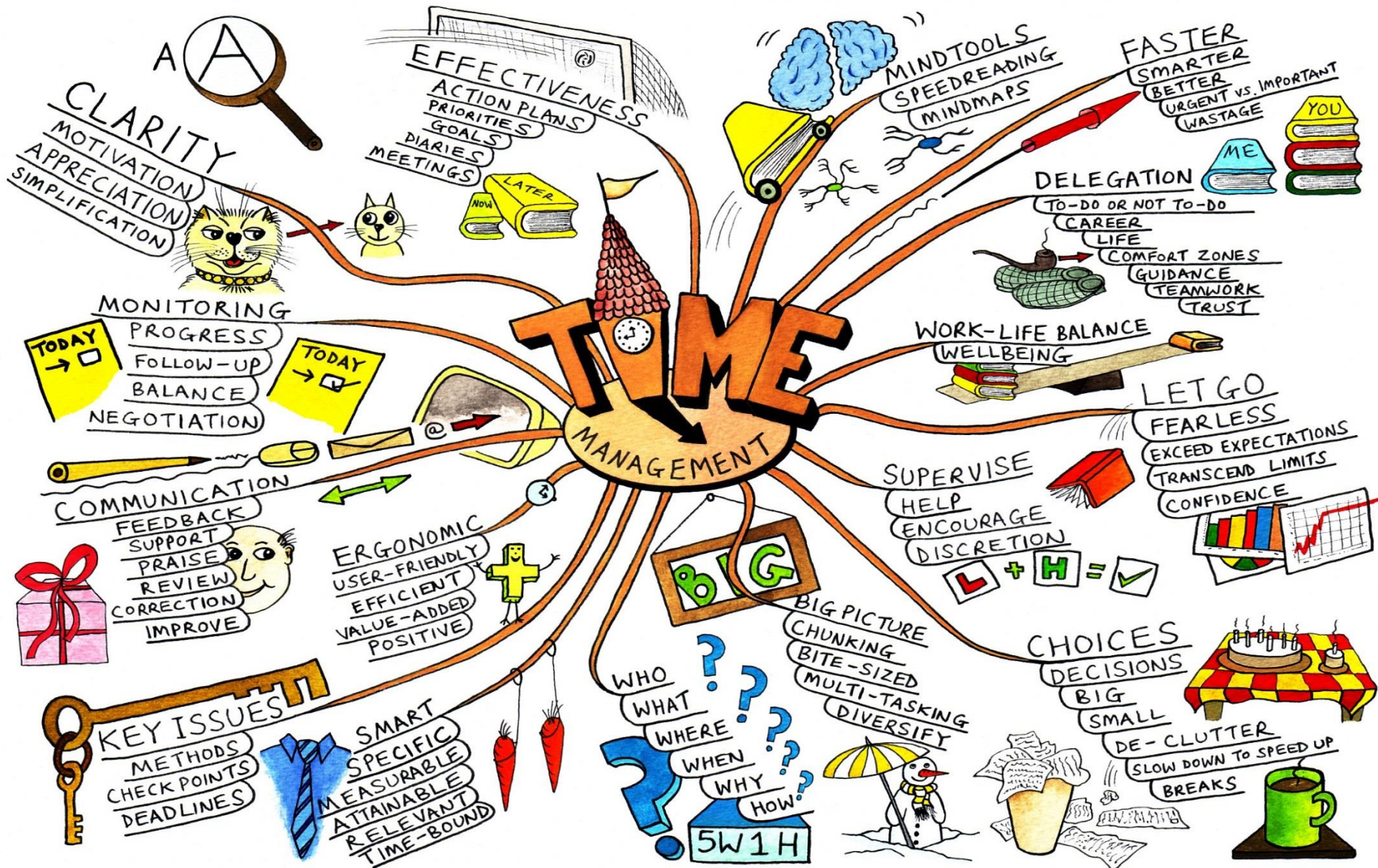
Example - Food



Mind Mapping Diagrams

- graphical representations of thought processes for brainstorming, problem solving, rational analysis, and decision marking.
- It includes thousands of ready-made graphics that you simply stamp to create your drawing

Mind Mapping Diagrams



Decision Matrix

- A Decision Matrix is a **structured tool used to compare multiple options** against a set of defined criteria. It helps you make **logical, objective decisions—especially when you have several alternatives and multiple factors to consider.**
- **Step-by-Step Process:**

1. List Your Options

What are you trying to choose between?

Example: Laptop A, Laptop B, Laptop C

2. Define the Criteria

What factors matter in your decision?

Example: Cost, Battery life, Speed, Weight

Decision Matrix

- **Step-by-Step Process:**

3. Assign Weights to Each Criterion

Give a weight based on how important each criterion is.

Weights usually add up to 1 or 100%.

Example: Cost → 30%Battery life → 20%Speed → 30%Weight → 20%

4. Rate Each Option Against Each Criterion

Use a consistent scale (e.g., 1 to 5 or 1 to 10), where a higher score means better performance.

5. Calculate the Weighted Scores

Multiply the score by the weight for each criterion.

Then add all weighted scores for each option.

Decision Matrix

Criteria	Weight	Laptop A	Laptop B	Laptop C
Cost	0.3	4	5	3
Battery Life	0.2	5	3	4
Speed	0.3	3	4	5
Weight	0.2	2	4	5
Total Score		3.5	4.2	4.1

Decision Matrix

			Alternatives		
		Wt	Vendor 1	Vendor 3	Vendor 4
Criteria	Cost	.30	4	4	4
	Response time	.17	3	3	5
	Training time	.17	2	4	5
	Ease of use	.17	1	4	4
	Strong team	.10	3	4	2
	Team experience	.10	3	4	2
Total		1.0	16	23	22
Weighted total			2.8	3.8	3.9

EXAMPLE DECISION MATRIX

A group is deciding between three options for new software. They have identified four criteria to rate each option on a 1-to-5 scale (5 = best). On the right, they've added weights to each criteria in order to prioritize certain criteria over others.

Decision Making Criteria	Option A	Option B	Option C
Cost	3	4	5
Features	3	3	5
User Friendly	5	2	4
Customer Support	5	4	2
Total	16	13	16

The decision matrix without weights (above) works well when all criteria are considered equally important.

Decision Making Criteria	Weight	Option A	Option B	Option C
Cost	4	3	4	5
Features	3	3	3	5
User Friendly	2	5	2	4
Customer Support	1	5	4	2
Weighted Total		36	33	45
Rank		2	3	1

Decision matrix

DESIGN CRITERIA WEIGHTING FACTOR	USE OF STD. PARTS	SAFETY	SIMPLICITY & MAINT.	DURABILITY	PUBLIC ACCEPT.	FEASIBILITY	COST TO DEVELOP	COST TO OVEN.	APPEARANCE	SUM
ALTERNATIVES	0.08	0.12	0.10	0.10	0.10	0.20	0.03	0.07	0.05	1.0
LOOSE BULKER	90 7.2	90 10.8	90 9.0	90 9.0	65 6.5	70 14.0	90 2.7	90 3.6	90 4.5	79.0
VACUUM COLLECTOR	80 6.4	90 10.8	90 9.0	90 9.0	60 6.0	70 14.0	50 1.5	70 3.2	80 4.0	75.7
SHREDDER	50 4.0	60 7.2	70 7.0	70 7.0	85 8.5	65 13.0	35 1.7	60 2.4	45 2.25	65.9
CHEMICAL DECOMPOSER	50 4.0	35 4.2	80 8.0	60 6.0	30 3.0	90 18.0	55 1.7	50 2.0	70 3.5	68.8
COLLECTOR - POWER-LINE			OUT DUE TO FEASIBILITY							
PORTABLE INCINERATOR			OUT DUE TO FEASIBILITY							